

**School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi-110067
Database Management Systems (CS-407)
End Semester Exam, Winter Semester – 2024**

Duration: 3 Hours

Max. Marks: 50

All questions are compulsory and carry equal marks:

Q1 (a) What is lock and explain Two-phase locking protocol.

Q1 (b). Given two sets of FDs for a relational variable R {A, B, C, D, E}, Are they equivalent?

1. $A \rightarrow B, AB \rightarrow C, D \rightarrow AC, D \rightarrow E$

2. $A \rightarrow BC, D \rightarrow AE$.

Q2 (a). Add *lock* and *unlock* instructions to the following transactions T1 and T2, so that they observe the *two-phase locking* protocol.

T1: read(A), read (B), if A = 0 then B=B+10, write (B)

T2: read (B), read (A), if B=0 then A=A+10, write (A)

Q2 (b). Consider the following schedule S: T1: Read(A), T1: Write(A), T2: Read(A), T1: Read(B), T2: Write(A), T1: Write(B), T2: Read(B), T2: Write(B). Show conflict instructions (if any) in the given schedule. Draw an equivalent conflict serializable schedule.

Q3 (a). Consider a relation R (U, V, W, X, Y, Z) with MVDs $UW \twoheadrightarrow XZ$ and $V \twoheadrightarrow Y$. Decompose the relation R into 4NF relations.

Q3 (b). Differentiate between *wait-die* and *wound-wait* deadlock preventions techniques.

Q4 (a) What is serializability? Differentiate between conflict and view serializability. (2)

Q4 (b). Consider the relations and write the relational algebra expressions of the queries: (8)

employee (person-name, street, city)
works (person-name, company-name, salary)
company (company-name, city)
manages (person-name, manager-name)

- (i) Find the names of all employees who live in the same city and on the same street as do their managers.
- (ii) Find the names of all employees in this database who do not work for First Bank Corporation.
- (iii) Find the names of all employees who earn more than every employee of Small Bank Corporation.
- (iv) Assume the companies may be located in several cities. Find all companies located in every city in which Small Bank Corporation is located.

Q5 Consider the relations and write the SQL expressions of the queries: (10)

Sailors (SailorId, SailorName, Rating, Age)
Reserves (SailorId, BoatId, Day)
Boats (BoatId, BoatName, Color)

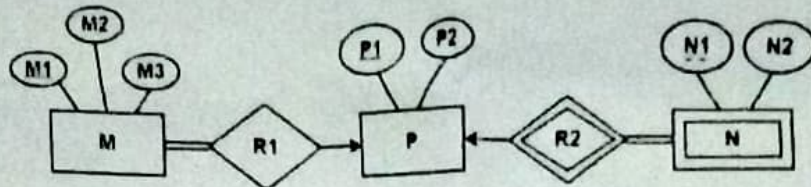
- a) Find the names of sailors who have reserved a red or a green boat.
- b) Find the names of sailors who have reserved at least two boats.
- c) Find the names of sailors who have reserved all boats.
- d) Find the names of boats that are reserved by more than one sailor.

School of Computer and Systems Sciences
Jawaharlal Nehru University
Mid Semester Examination, (Monsoon Semester 2024)
Database Management Systems (CS-706)

Duration: 90 Minutes

Max. Marks 25

Q1 (a). Map the following E-R diagram into an equivalent relational schema with minimum number of relations.



(2)

(b) Consider the relation schema $R = (E, F, G, H, I, J, K, L, M, N)$ and the set of functional dependencies: $EF \rightarrow G, F \rightarrow I, J, EH \rightarrow K, L, K \rightarrow M, L \rightarrow N$

What is the key for R?

(2)

(c) What is aggregation? Give two suitable examples where this concept can be useful.

(2)

(d) Define the terms primary key, candidate key, super key and foreign key.

(2)

Q2 Consider a relation schema $R = \{A, B, C, D\}$ with functional dependencies

$A \rightarrow B, B \rightarrow C, C \rightarrow D, \text{ and } D \rightarrow B.$

Determine whether the decomposition of R into $R_1(A, B), R_2(B, C)$ and $R_3(B, D)$ is lossy or lossless.

(4)

Q3. For a relation schema $R = (A, B, C, D)$ the following set of functional dependencies hold:

(4)

$B \rightarrow A, AD \rightarrow BC \text{ and } C \rightarrow ABD.$ Determine the canonical cover for this set of FDs.

Q4. Let suppose, a relation $R(P, Q, R, S)$ with a set of Functional Dependency $FD = (PQ \rightarrow R, R \rightarrow S, S \rightarrow P)$ is decomposed in $R_1(P, Q, R)$ and $R_2(R, S)$. Determine whether the decomposition is dependency preserving or not.

(4)

Q5. For the following problem definition, draw an E-R diagram with appropriate mapping cardinalities and identifying attribute/s

ABC prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, ABC relies on a company-wide information system. Shipped items are the heart of the ABC product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the ABC system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard ABC transportation events (i.e., flights, truck deliveries). These transportation events are characterized by a unique scheduleNumber, a type (e.g. flight, truck), and a deliveryRoute.

(5)

60